CUSTOMER PURCHASE ANALYSIS

End-to-End Data Analytics Project Report

Python | SQL | Power BI

Project Domain	**Retail / Customer Analytics**
Prepared by	Padmanaban L
Tools	Python, Pandas, SQL, Power BI
Status	Completed

This report documents customer purchase behavior, sales performance, and customer characteristics using Python data preparation and exploratory analysis, SQL business queries, and interactive Power BI dashboards.

Executive Summary

The Customer Purchase Analysis project examines purchase amount, product categories, customer demographics, shopping frequency, subscription status, discounts, promo-code usage, shipping methods, payment methods, and seasonal sales patterns. Python is used to inspect and clean the customer dataset, impute missing review ratings using category-level medians, and perform exploratory analysis. SQL is used to answer 30 business questions. Power BI presents a Sales Overview and Customer Analysis dashboard.

The supplied dashboard displays approximately 3.9K customer records, total purchase amount of 233K USD, average review rating of 3.75, and promo-code usage of approximately 2K. Category sales shown are Clothing 104K, Accessories 74K, Footwear 36K, and Outerwear 19K. These are rounded dashboard labels.

1. Business Problem

Retail businesses need to understand which product categories and locations contribute most to sales, how customers purchase, and whether shopping behaviors differ by gender, age group, subscription status, discounts, promo codes, and purchase frequency. This project consolidates these indicators into an analytical workflow and interactive dashboard.

2. Project Objectives

- Measure total records, total sales, average purchase amount, and average review rating.
- Compare sales and average purchase amounts across categories, products, genders, and locations.
- Explore subscription, discount, and promo-code behavior.
- Analyze payment and shipping method popularity and sales contribution.
- Understand purchase frequency, repeat-customer behavior, and customer value segments.
- Build interactive Power BI dashboards to support business exploration.

3. Dataset Overview

The Python script reads customer_history (1).csv; the SQL script queries the customer_history table. Referenced fields include Customer ID, Age, Gender, Category, Item Purchased, Purchase Amount (USD), Review Rating, Subscription Status, Discount Applied, Promo Code Used, Payment Method, Shipping Type, Frequency of Purchases, Previous Purchases, Location, and Season.

The raw CSV and executed SQL result tables were not supplied. Therefore, this report uses only the values visible in the supplied dashboard screenshots and does not invent row-level outputs.

4. Tools & Technologies

Tool	Purpose
Python / Pandas	Dataset loading, inspection, missing-value treatment, categorical validation, EDA, CSV export
Matplotlib	Visualizing customer distribution and category sales
SQL	Business analysis using aggregates, grouping, filtering, sorting, CASE, and subqueries
Power BI	Interactive KPI cards, category slicers, charts, and customer analysis
Google Colab	Notebook execution and cleaned-file download

5. Methodology

1. Load the customer dataset into a Pandas DataFrame and inspect sample rows, schema, and descriptive statistics.
2. Check missing values, duplicate rows, data types, review-rating statistics, and unique categorical values.
3. Impute missing Review Rating values using the median within each Category.
4. Perform exploratory analysis of gender distribution, category sales, top products, and subscription sales.
5. Export the prepared data as customer_history_cleaned.csv.
6. Use SQL queries to answer 30 business questions covering sales, customer segments, purchase behavior, and operational preferences.
7. Build and review Power BI visuals and dashboard KPIs.

6. Python Data Cleaning & Analysis

Dataset loading: The script imports pandas and reads customer_history (1).csv into df.

Initial inspection: head(), info(), describe(), describe(include='object'), and describe(include='all') are used to inspect the dataset.

Data-quality checks: isnull().sum(), duplicated().sum(), dtypes, unique values, and value_counts() are used to review quality and categories.

Missing review-rating treatment: Missing Review Rating values are filled with the median Review Rating within each Category using groupby('Category').transform(...).

Gender analysis: The script calculates gender counts and percentages and plots customer distribution by gender.

Category sales: Purchase Amount (USD) is summed by Category, sorted descending, and visualized.

Product sales: Sales are aggregated by Item Purchased; the code plots the top 10 products.

Subscription analysis: Purchase amount is grouped by Subscription Status and summarized by count, sum, and mean.

Cleaned-data export: The prepared DataFrame is saved as customer_history_cleaned.csv and downloaded from Google Colab.

Python code highlights

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(lambda x: x.fillna(x.median()))
category_sales = df.groupby('Category')['Purchase Amount (USD)'].sum().sort_values(ascending=False)
subscription_sales = df.groupby('Subscription Status')['Purchase Amount (USD)'].agg(['count','sum','mean'])
df.to_csv('customer_history_cleaned.csv', index=False)
```

7. SQL Business Analysis

The supplied SQL file contains 30 commented business questions. The analysis is organized into the following themes:

Sales performance: Total sales, average purchase, highest-sales categories, average category purchase, category ratings, best-selling and lowest-selling products.

Customer demographics: Sales and average purchase by gender; average customer age; sales by location and top locations.

Subscription and promotion: Sales and average purchase by subscription status; discount comparisons; promo-code usage and spending.

Payment and shipping: Most-used and highest-sales payment methods; most popular and highest-sales shipping types.

Purchase behavior: Sales by purchase frequency; customers above average purchase; highest-spending customer; high-, medium-, and low-value transaction labels.

Customer loyalty and segmentation: Repeat versus one-time customer counts and average purchase; repeat-customer transaction percentage; high-value transaction count; average purchase for customers aged 40+.

SQL techniques used

- COUNT(), SUM(), and AVG() for volume and sales metrics.
- ROUND() for presentation-friendly numeric values.
- GROUP BY and ORDER BY for segment comparisons and rankings.
- LIMIT for top/bottom result lists.
- CASE expressions for customer value and repeat-customer segments.
- Scalar subquery to compare purchases against the overall average.

The SQL file contains query definitions rather than saved execution outputs. Add verified query result tables after running the script against the database.

8. Power BI Dashboard

The supplied Power BI report contains two dashboard views. Category slicers allow users to explore Accessories, Clothing, Footwear, and Outerwear.

Page 1 — Customer Purchase Analysis: Sales Overview

The Sales Overview displays customer count, total purchase amount, and average review rating. Visuals include purchase amount by category and location, purchase trend by season, and purchase amount by gender.

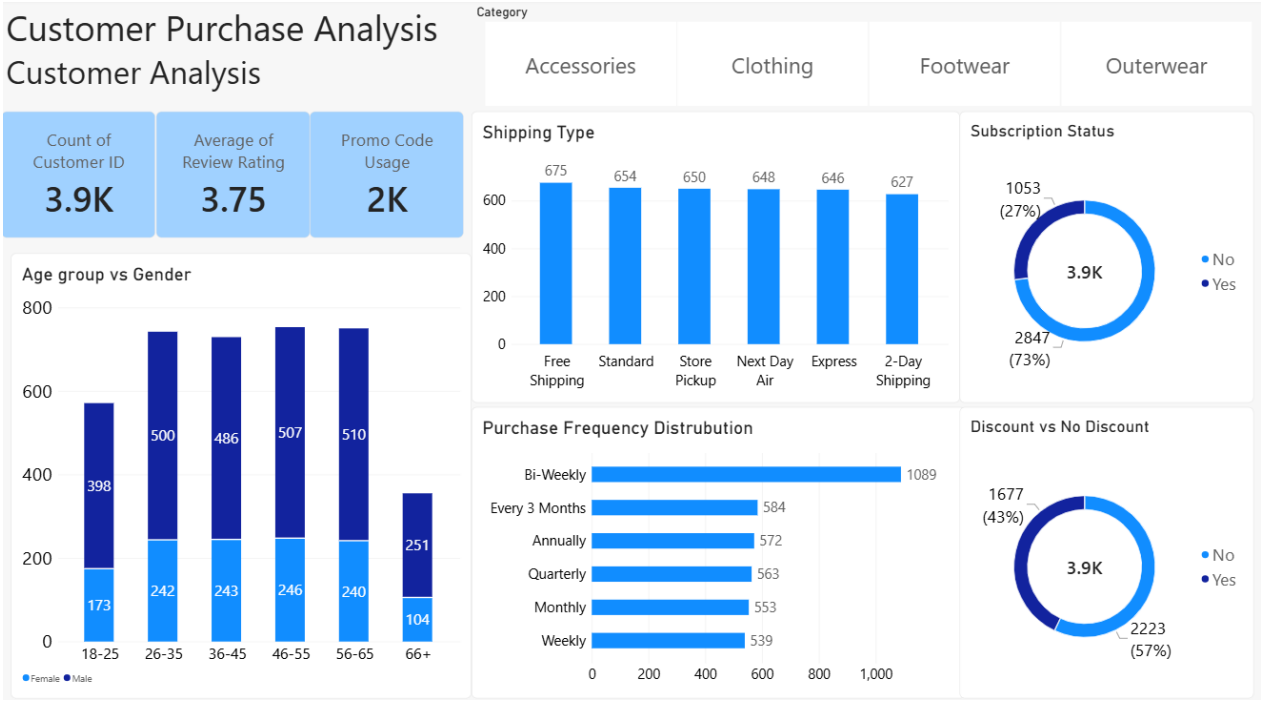


Figure 1. Customer Purchase Analysis — Sales Overview.

Page 2 — Customer Purchase Analysis: Customer Analysis

The Customer Analysis page includes customer count, average review rating, promo-code usage, age group versus gender, shipping type, subscription status, purchase frequency distribution, and discount versus no-discount.

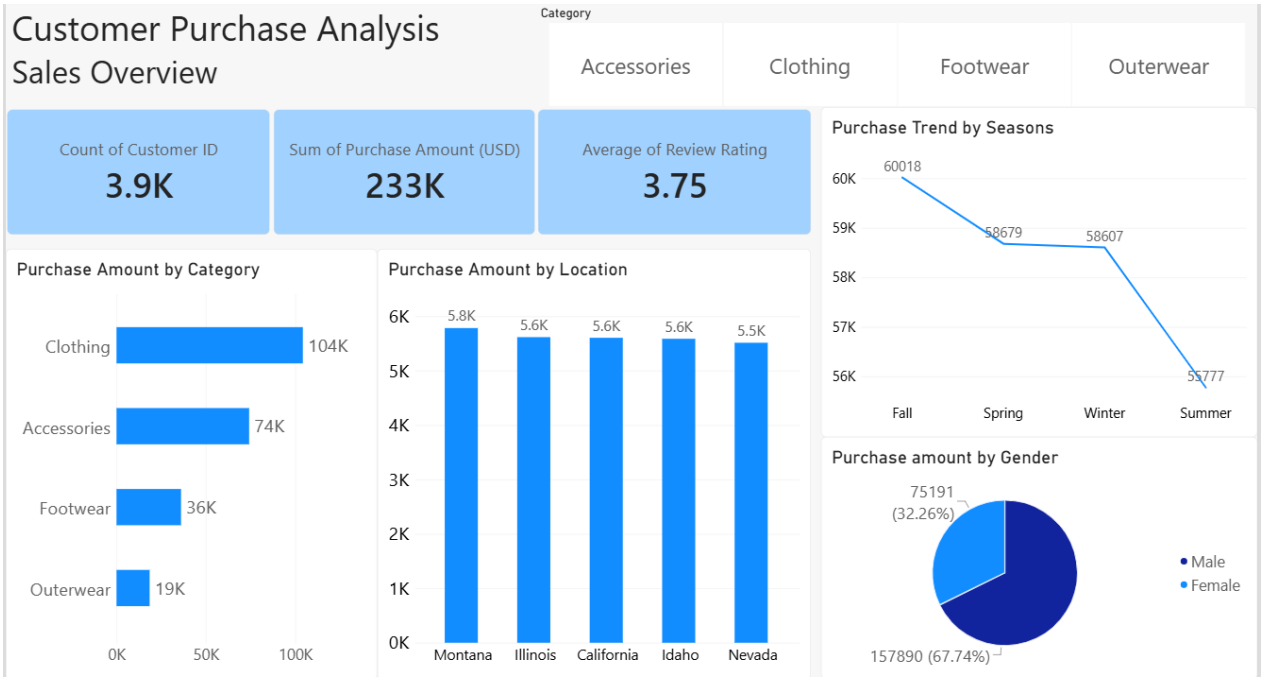


Figure 2. Customer Purchase Analysis — Customer Analysis.

9. Dashboard KPI & Sales Summary

Metric	Displayed value
Customer records	3.9K (rounded)
Total purchase amount	233K USD (rounded)
Average review rating	3.75
Promo-code usage	2K (rounded)
Clothing sales	104K USD (rounded)
Accessories sales	74K USD (rounded)
Footwear sales	36K USD (rounded)
Outerwear sales	19K USD (rounded)

10. Key Observations

- The dashboard displays total purchase amount of approximately 233K USD and an average review rating of 3.75.
- Clothing is the highest displayed category by sales at approximately 104K USD, followed by Accessories at 74K USD.
- The seasonal chart displays Fall at 60,018, Spring at 58,679, Winter at 58,607, and Summer at 55,777.
- The Customer Analysis page displays subscription status as 2,847 No (73%) and 1,053 Yes (27%).
- Purchase frequency chart shows Bi-Weekly as the largest displayed group at 1,089.
- The gender purchase chart displays Male 157,890 (67.74%) and Female 75,191 (32.26%).

These observations are descriptive and based on rounded screenshot labels. They do not establish that a promotion, subscription, or demographic factor causes higher spending.

11. Business Recommendations

- Review product availability, merchandising, and campaign opportunities for high-sales categories such as Clothing and Accessories.
- Investigate lower-sales categories and products before making inventory or promotion decisions.

- Compare average purchase amount—not only total sales—across customer segments to avoid volume-only conclusions.
- Test subscription and promo-code campaigns using controlled comparisons and track conversion, average order value, and repeat purchases.
- Use customer feedback and review ratings to identify product or service improvement opportunities.
- Monitor sales by season, location, payment method, and shipping preference to support operational planning.
- Validate all dashboard measures and SQL outputs against the source dataset before presenting exact results.

12. Limitations

- Raw CSV and SQL execution outputs were not supplied for independent recalculation.
- Dashboard labels are rounded; some totals may not exactly match raw query outputs.
- The supplied SQL comments say 'top 10' in some questions while the query uses LIMIT 5; report interpretations should follow the actual query behavior.
- The analysis is descriptive and does not include statistical testing or causal inference.

13. Conclusion

The Customer Purchase Analysis project demonstrates a practical end-to-end analytics workflow. Python supports data inspection, review-rating imputation, exploratory analysis, and cleaned-data export. SQL structures business questions around sales, customer segments, and purchase behavior. Power BI communicates the findings through interactive sales and customer dashboards. Together, these tools provide a foundation for evidence-based retail analysis.

Appendix — Source files

Python script: sales_data_data_cleaning(1).py. SQL script: Sales Data Analysis(1).sql. Power BI file: Sales dataset project(1).pbix. Dataset referenced by the script: customer_history (1).csv.